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Microaerobic pretreatment of sewage sludge: Effect of air flow rate, pretreatment time and temperature on the aerobic process and methane generation

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ABSTRACT

The effect of air flow rate (AFR), pretreatment time, and temperature on a microaerobic pretreatment were determined, including its effect on CH_4 generation. For this, four AFR (0.1, 0.3, 0.5 and 1 vvm), three temperatures (20, 25 and 35 °C) and three aeration times (24, 48 and 60 h) were evaluated in 1-L and 5-L bioreactors. The effect on CH_4 production was evaluated in 280-mL bioreactors, using previously aerated sewage sludge. It was found that the micro-aerobic pretreatment improves the hydrolytic process, with an increase of proteins, total sugars, soluble chemical oxygen demand (COD), and a greater removal of volatile suspended solids (VSS) compared to a system without aeration. It was determined that an AFR = 0.3 vvm, 48 h of aeration time, and a temperature of 35 °C were the best operational conditions for increasing hydrolysis rate. VSS removal under the aerobic pretreatment follows first order kinetics. Finally, the aeration pretreatment allows a 211% increase of methane generation compared to a process without pretreatment, improving also total COD and VSS removal.

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1. Introduction

Anaerobic digestion (AD) of sewage sludge is a valuable treatment that results in the reduction of sludge volume, destruction of pathogenic organisms, stabilization of the sludge, and production of an energy-rich biogas (Appels et al., 2008). This anaerobic digestion involves a number of metabolic reactions including hydrolysis, fermentation, and methane formation. According to the literature (Zhu et al., 2009; Abelleira-Pereira et al., 2015), the main limiting factors of sewage sludge AD are associated with the hydrolysis step. Therefore, various sludge disintegration methods have been studied as a pretreatment, where slowly degradable particulate organic matter is converted into low molecular weight readily biodegradable compounds, bypassing the rate-limiting hydrolysis stage. The pretreatments that have been reported are physical-chemical and mechanical methods (Mata-Alvarez et al., 2000; Divyalakshmi et al., 2015), thermo-chemical treatment (Zhen et al., 2014; Kim et al., 2015), and biological methods (Saidu et al., 2014; Liu et al., 2015; Mutschlechner et al., 2015).

Among the biological methods, the use of micro-aeration as a treatment prior to anaerobic digestion has received less attention, even though it has been reported that the limited oxygen supply increased the enzymatic hydrolysis rates for the case of complex organic matter in batch tests (Johansen and Bakke, 2006). With limited oxygen, it has been suggested that the facultative anaerobes maintain a low redox potential and supply more growth factors for the strict anaerobes (Gerritse et al., 1990; Kato et al., 2005). These findings suggest that hydrolysis, and therefore COD solubilization, might be improved by micro-aeration. Some studies about the effect of micro-aeration on anaerobic digestion have been reported, mainly using municipal solid waste (MSW) as substrate (Charles et al., 2009; Nguyen et al., 2007; Xu et al., 2014; Fu et al., 2015), indicating that micro-aeration has an important effect in terms of enhancing the anaerobic digestion process and this needs further investigation. Other studies using vegetable and flower wastes (Zhu et al., 2009) and grass silage (Jagadabhi et al., 2010) have also been carried out, indicating that the amount of oxygen supplied and the micro-aeration time needs to be further optimized. Mshandete et al. (2005) studied the aerobic pretreatment of sisal pulp in mesophilic anaerobic digestion using an mixed culture of activated sludge. They indicated that the correlation between high enzyme

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activity and high methane yield using an aerobic pretreatment suggests that such a short pretreatment period may be an alternative option for increasing solubilization of sisal pulp and promoting methane productivity.

Regarding the effect of micro-aeration on the AD of sewage sludge, few studies have been reported. Jenicek et al. (2011), using a continuous bioreactor and pumping air continuously into the sludge recirculation stream, showed that the VSS/TSS ratio of the digested sludge, soluble COD concentration, ammonia nitrogen, and phosphate concentration decreases in all experiments under micro-aerobic conditions. Jenicek et al. (2010), working with a continuous stirred tank reactor (CSTR) with continuous micro-aeration, indicated that efficient H_2S removal, higher specific methane production, lower methane concentration in the biogas due to dilution with nitrogen remained from the dosage of air, better sludge liquor quality, and lower foaming potential and foam stability were obtained using micro-aeration. Dumas et al. (2010) studied the efficiency of a hyper-thermophilic (65 °C) aerobic process coupled with a mesophilic (35 °C) digester, getting a 30% increase of COD removal and 20%–40% of biodegradability. However, the methanization yield of the process was similar to that of a conventional process, probably due to the uptake of carbon from the facultative biomass. Ramos and Fdz-Polanco (2013) performed anaerobic digestion in a continuous pilot reactor in order to see the effect of microaeration on the performance of anaerobic digestion under overload conditions, showing that oxygen seemed to form a more stable digestion system, which meant increased ability to deal successfully with overloads. Therefore, none of these studies had reported the effect of parameters such as pretreatment time, temperature, or air flow rate (AFR) on pretreatment. Also, the kinetics of the hydrolytic process in the aerobic pretreatment has been scarcely reported. Knowledge of the effect of these parameters on the pretreatment can help to determine an operation range within which both pretreatment and anaerobic digestion can be operated efficiently. Furthermore, the kinetics of the hydrolytic process can help in understand and optimiz the aerobic pretreatment.

Therefore, the goal of the present work was to determine the effect of AFR and aeration time and temperature on the pre-aeration process, including also its effect on the anaerobic digestion process. The kinetics of VSS degradation in the pre-aeration process was also determined.

2. Materials and methods

2.1. Substrate, inocula and experimental design

The substrate for all the assays was a mixture of primary (60%) and secondary (40%) sludge from “La Farfana”, a municipal wastewater treatment plant operated by Aguas Andinas in Santiago, Chile. The inoculum had a specific methanogenic activity of 0.32 g CH_4 g^{-1}VSS d^{-1}, and it was obtained from an anaerobic reactor of La Farfana.

Five assays were performed at the laboratory scale under the conditions presented in Table 1. Each assay was ran on three samples of mixed sludge taken at different times, and they were run in duplicate. The goal of assay 1 was to know the influence of the micro-aeration pretreatment on the hydrolysis process. Assay 2 was meant to study the effect of different pretreatment times (24, 48 and 60 h) on the solubilization of sewage sludge. This assay also measured protein and total sugar in order to evaluate the solubilization grade. Assay 3 was carried out in order to determine the influence of temperature on hydrolysis in the microaerobic pretreatment. Assay 4 was run in order to determine the kinetics of hydrolysis in the micro-aerobic pretreatment. Finally, assay 5 was carried out to determine the effect of the micro-aerobic pretreatment on CH_4 generation; in this case, the contents of the aerobic reactors were taken and put in several 280-mL anaerobic bioreactors (200 mL of hydrolyzed sewage sludge plus 50 mL of inoculum), similar to the system used by Huilinir et al. (2015, 2014).

The kinetics of hydrolysis (Assay 4) was studied by the integral method using first order kinetics, since this is followed by the most widely used hydrolysis model (Vavilin et al., 2008). The first order kinetics indicates that

$$\frac{d(VSS)}{dt} = -k \cdot VSS \tag{1.1}$$

where VSS is the VSS concentration (g/L) and k (h^{-1}) is the kinetic constant. Integrating, we get:

$$\ln(VSS)_t = -k \cdot t + \ln(VSS)_0 \tag{1.2}$$

Equation (1.2) is a straight line and k can be calculated from its slope. Linearization was performed in *Microsoft Excel 2010*.

2.2. Experimental setup

Assays 1, 2 and 3 were run in 1-L glass reactors, operating with an effective volume of 900 mL. Assays 4 and 5 were ran in bigger reactors, with a total volume of 5 L and an operational volume of 4 L. In each pretreatment, air was injected continuously into the bioreactors by means of mini-compressors (0–10 Lpm) and distributed by small diffusers. The air flow rate (AFR) was calibrated before the pretreatment assays using the volume variation method. As shown in Table 1, assays 1 and 2 were carried out at room temperature without temperature contro, while assays 3, 4 and 5 were run with temperature control.

2.3. Chemical and statistical analysis

The following parameters were determined: total COD (t_{COD}), soluble COD (s_{COD}), total and suspended solids, volatile suspended solids, pH, ammonia, proteins, and total sugars. COD (total and soluble) was measured by the colorimetric method according to (APHA et al., 2012). Total and suspended solids, volatile suspended solids, ammonia and pH were measured according to APHA et al. (2012). Proteins were measured by the Bradford method (Kruger, 1994), and total sugars were measured by the colorimetric method (Dubois et al., 1956).

Methane production was measured by volumetric displacement, connecting an inverted falcon tube containing a 3% w/w NaOH solution in order to eliminate CO_2 and H_2S as the main impurities from the biogas, displacing only the methane volume. The resulting biogas travels through the flexible tubing to the Falcon tube, where it comes in contact with the NaOH solution, forming sodium carbonate and bubbling only methane. The methane collected in the Falcon tube was sucked with a syringe and released in a flame to observe its presence.

The Minitab 8 software was used for the statistical processing and analysis of the data. An ANOVA analysis was performed and a comparison of 95% confidence intervals for mean values was made.

3. Results and discussion

3.1. Effect of aeration flow rate (AFR) on hydrolysis (Assay 1)

The effect of different AFR on VSS, s_{COD} and pH is shown in Table 2. No significant differences in the behavior of duplicate reactors in all assays were seen. Therefore, only the results of one of the duplicate reactors will be reported. It is seen that VSS decreased

Table 1
Experimental conditions of the assays at laboratory scale.

Assay number	Aeration level (vvm)[a]	Aeration period (hours)	Operating temperature (°C)	Batch anaerobic digestion at 35 °C
1	0	48	18–20[b]	NO
	0.1	48	18–20[b]	NO
	0.3	48	18–20[b]	NO
	0.5	48	18–20[b]	NO
	1.0	48	18–20[b]	NO
2	0.3	24-48-60	18–20[b]	NO
3	0.3	48	18–20[b]	NO
	0.3	48	25	NO
	0.3	48	30	NO
	0.3	48	35	NO
4	0.3	48	25	NO
	0.3	48	35	NO
5	0	48	35	Yes
	0.3	48	35	Yes

[a] vvm: air volume perliquid volume per minute.
[b] Ambient temperature.

Table 2
Effect of pre-aeration on VSS concentration, soluble COD concentration, and pH after 48 h of aeration.

Aeration rate vvm	Initial VSS mg L^{-1}	Final VSS mg L^{-1}	% de-crease	Initial s_{COD}, mg L^{-1}	Final s_{COD}, mg L^{-1}	% in-crease	Initial pH	Final pH	% in-crease
0	3240 ± 120	3000 ± 60	7.4	116.5 ± 0.0	128.1 ± 0.0	10.0	7.46	7.65	2.54
0.1	3130 ± 90	2690 ± 40	14.1	110.6 ± 17.5	139.7 ± 11.6	26.3	7.57	7.82	3.30
0.3	3760 ± 140	2400 ± 50	36.2	93.2 ± 11.6	172.4 ± 4.7	85.0	7.61	7.92	4.07
0.5	3450 ± 130	2530 ± 50	26.7	104.8 ± 11.6	153.7 ± 4.7	46.7	7.53	7.81	3.72
1	3530 ± 110	2810 ± 60	20.4	99.0 ± 5.8	128.1 ± 7.0	29.4	7.44	7.73	3.89

in all the assays, reaching a maximum of 37% in 2 days at vvm = 0.3. The lowest VSS decrease occurred in the anaerobic assay (without aeration), with only 7% removal. This difference between the process with and without pre-aeration shows that the hydrolysis under the aerobic process is faster than the anaerobic one, with a higher degree of VSS destruction (Lim and Wang, 2013). This also accounts for the higher production of S_{COD} under aerobic conditions (Table 2), due to the solubilization of the complex organic matter (Burgess and Pletschke, 2008). In fact, the S_{COD} increase under aerobic conditions varied between 26% and 85%, while it was only 10% under anaerobic conditions. This behavior agrees with the results presented by Lim and Wang (2013) and Jang et al. (2014), but it disagrees with other reports, which show that soluble COD in pre-aerated reactors decreases with respect to the anaerobic process (Jenicek et al., 2011; Diak et al., 2013). This difference can be accounted for because the pre-aerated system used here is an extra step, where the sludge was aerated uninterruptedly during 48 h, and after that it was digested anaerobically; in contrast, Jenicek et al. (2011) and Diak et al. (2013) aerated the system *in-situ* (in the same bioreactor and jointly with the anaerobic digestion).

Regarding pH, all the assays increased the pH value, with the reactors with pre-aeration again undergoing a larger increase, between 3.3% and 4.07%. This pH increase can be explained by the generation of ammonia in the hydrolysis of protein (Huilinir et al., 2011). The pH increase does not agree with the behavior reported by Lim and Wang (2013), who showed that pH decreased during the pre-aeration step (4 days). In our case, the presence of a high nitrogen content in the sludge allows an increase in pH and better operational conditions for the anaerobic step.

It should be pointed out that from 24 h after the beginning of the assays, H_2S concentration in the liquid phase was undetectable. This behavior is expected because of the oxidation of sulfur by sulfur-oxidizing biomass, a situation also reported in other studies (Krayzelova et al., 2014). Furthermore, the oxygen concentration in all the assays also had values lower than the lowest limit of the sensor, showing that the conditions in the assays correspond to a micro-aerobic system.

Therefore, from these results it can be stated that pre-aeration allows increasing the hydrolysis process, with an AFR = 0.3 vvm as the best option in order to increase the hydrolysis rate of complex organic matter.

3.2. *Effect of aeration time on protein and carbohydrates hydrolysis (Assay 2)*

Assay 2 was used to determine the effect of different pre-aeration times on the process at 0.3 vvm. The effect was seen in the protein and total sugar behavior, also including total ammonia nitrogen (TAN). Fig. 1A shows that the soluble protein concentration increased with respect to the anaerobic test. The largest increase (341%) was achieved at 48 h, while the lowest (42%) occurred at 60 h. This behavior can be accounted for by the hydrolysis of particulate proteins, which release amino acids and ammonia (Fig. 1C). The increase can also be accounted for because a part of the mixed sewage sludge coming from an activated sludge process (40%) developed flocs in order to improve the decantation. These flocs can release the soluble protein in the liquid phase, a process called “deflocculation”, and it is attributed mainly to a low aeration in the activated sludge process. This was exactly what happened in the pre-aeration treatment, so this can also explain the larger production of soluble protein (Morgan-Sagastume and Allen, 2005).

Fig. 1B shows the total sugar concentrations at different aeration times, and it is seen that longer aeration times increase the soluble sugars in the liquid phase, with increases up to 300% at 60 h. Since it is known that hydrolysis is the rate-limiting step for carbohydrate conversion (Miron et al., 2000), the results show that the pre-aeration improves carbohydrate hydrolysis, allowing better solubilization for further anaerobic processes. Fig. 1B also shows that the difference between the soluble sugars obtained at 48 h and 60 h is small, so the improvement of carbohydrate hydrolysis is marginal when aeration time increases from 48 h to 60 h.

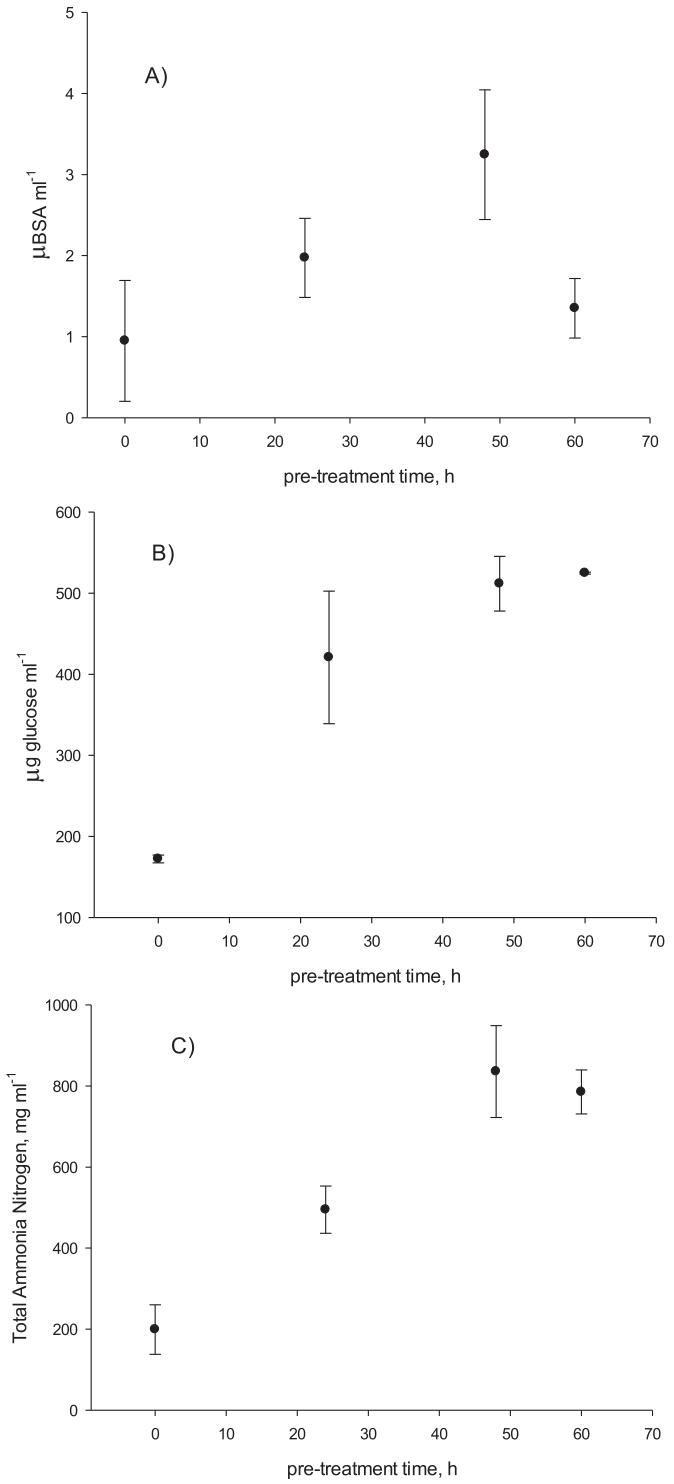


Fig. 1. Values of protein, total sugars and ammonia at the end of different pre-aeration times: A) protein (measured as bovine serum albumin equivalent); B) total sugars (measured as glucose equivalent); C) ammonia.

Fig. 1C shows the total ammonia nitrogen (TAN) concentration obtained at different pretreatment times. It is seen that aeration time increases the release of ammonia in the liquid phase mainly because of protein hydrolysis. The same asa in assay 1. this increase also accounts for the pH increase in the media. In this assay pH increases from 5.6 to 6.4, a situation that also favors the further anaerobic process. It is important to note that ammonia nitrogen concentration does not rise to inhibitory values for anaerobic digestion, varying between 1500 and 3000 mg TAN L^{-1} (Chen et al., 2008; Montalvo et al., 2014), although at times it has been possible to acclimatise anaerobic systems at as much as 4000 mg L^{-1} of TAN (Gao et al., 2015).

All earlier results show that a pre-aeration time of 48 h is enough to increase the hydrolysis of proteins and carbohydrates.

3.3. Effect of temperature on aerobic hydrolysis (Assay 3)

The effect of temperature on the microaeration pretreatment was studied in assay 3. The results of soluble COD are given in Table 3, where it is seen that a temperature increase causes increased soluble COD production, reaching as much as 85% at a temperature of 35 °C. This shows that the hydrolysis was favored at higher temperatures, even though the solubility of oxygen decreases. As mentioned by Luo et al. (2012), the increase of soluble organic matter at higher temperatures is due mainly to a hydrolytic process, because the particles obtained have higher kinetic energy and facilitate enzyme–substrate interactions, thereby increasing the hydrolysis rate.

The effect of temperature on VSS degradation was also studied, and the results are shown in Table 4. The same as for soluble COD, VSS solubilization increases at higher temperatures, with a 42%increase at 35 °C. These results agree with the study of Komemoto et al. (2009), who show that the solubilization of complex organic matter (food waste) increases in a mesophilic temperature range. This also accounts for the greater presence of soluble CODs at 35 °C.

Therefore, according to these results it is clear that mesophilic temperatures allow good performance of aeration pretreatment.

3.4. Kinetics of the hydrolysis of VSS under aerobic pretreatment (Assay 4)

Once the effect of AFR, aeration time, and temperature were evaluated, the kinetics of hydrolysis under microaerobic conditions was studied at 25 °C and 35 °C. The experimental and model results are shown in Fig. 2, where it is seen that in both experiments the behavior of VSS solubilization follows first order kinetics, with values of R^2 greater than 0.95. First order kinetics has been established in several reports as a good model to represents the hydrolysis under both aerobic and anaerobic conditions (Vavilin et al., 2008; Batstone et al., 2010). It is also seen that solubilization was faster at 35 °C, as mentioned earlier, with values of 0.226 d^{-1} and 0.269 d^{-1} for k at 25 °C and 35 °C, which are in the range reported in the literature for anaerobic hydrolysis (Vavilin et al., 2008), and they also agree with reports that state that even though hydrolysis under aerobic conditions is faster, enzymatic activity under aerobic and anaerobic conditions is not different (Goel et al., 1998).

3.5. Effect of micro-aerobic pretreatment on organic matter removal and methane generation in an anaerobic process (Assay 5)

The effect of aeration pretreatment on methane generation and organic matter removal was studied in assay 5, which was performed with sewage sludge pretreated at 35 °C and 0.3 vvm for 48 h, and the results are presented in Figs. 3 and 4. Regarding total COD (Fig. 3A), it is seen that its total removal in the process with pre-aeration is 80%, while without aeration it is only 57%. This situation clearly shows that a pretreatment allows greater total COD removal, almost twice as great when the aeration pretreatment was applied. It should be mentioned that the original total COD of mixed sludge before aeration was 134.077 g L^{-1}, with a total removal of 84% with the pre-aeration anaerobic digestion in series, and 57% using only anaerobic digestion. A possible

Table 3
Soluble COD concentrations at the beginning and the end of the aeration pretreatment and at different temperatures.

Temperature [°C]	Initial soluble COD, (t = 0 h) [mg O_2 L^{-1}]	Final soluble COD (t = 48 h) [mg O_2 L^{-1}]	% Increase
Ambient	95.5 ± 2.3	107.1 ± 4.7	12.2
25	93.2 ± 4.7	116.5 ± 4.7	25.0
30	107.1 ± 4.7	153.7 ± 4.7	43.5
35	107.1 ± 4.7	195.6 ± 9.3	82.6

Table 4
VSS concentrations at the beginning and the end of the aeration pretreatment at different temperatures.

Temperature [°C]	Initial VSS (t = 0 h) [mg L^1]	Final VSS (t = 48 h) [mg L^{-1}]	% Decrease
Ambient	3100 ± 260	2430 ± 110	21.6
25	3000 ± 100	2200 ± 80	26.7
30	2900 ± 100	1930 ± 10	33.4
35	3250 ± 90	1870 ± 70	42.5

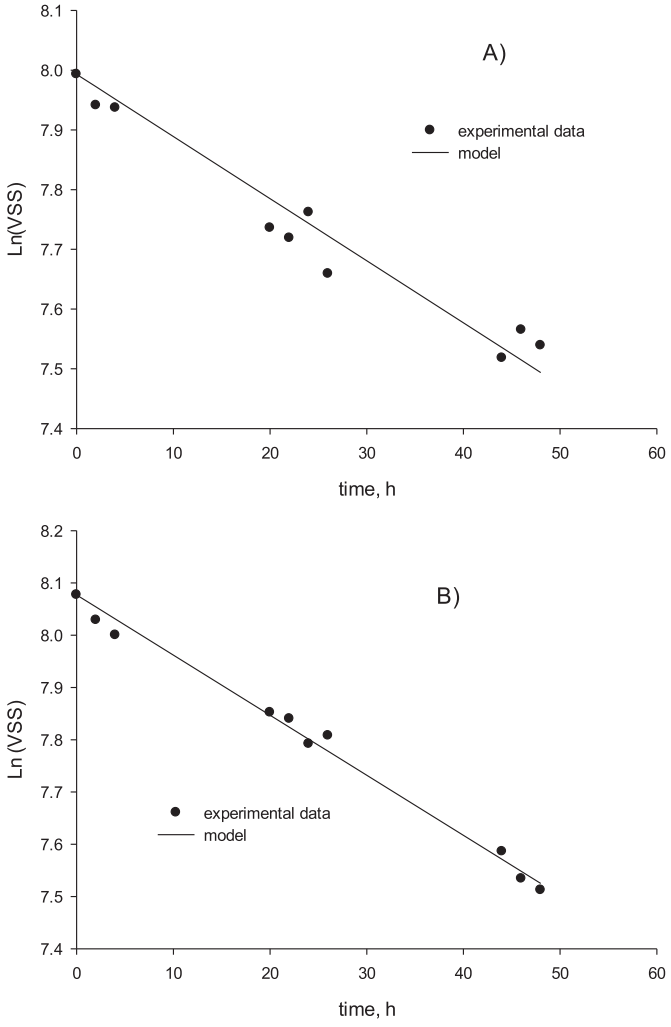


Fig. 2. Kinetics of aerobic hydrolysis at different temperatures. A) T = 25 °C; B) T = 35 °C.

explanation of this behavior is that the pre-aeration process improves the solubility of organic matter, making the rapidly biodegradable organic matter available to the fermentative and methanogenic biomass. Similar results were presented by Diak et al. (2013), who worked with primary sludge, and by Lim and Wang (2013), who worked with brown water and food waste.

Fig. 3B gives the results for soluble COD, showing that the trend in both reactors was similar, with an increase of soluble COD concentration until day 16, followed by its decrease. This trend was also found by Diak et al. (2013) using primary sludge pre-treated with micro-aeration. The soluble COD increase during the first 16 days can be attributed to the hydrolytic process, which generates soluble COD that later will be used in the methanogenesis (Burgess and Pletschke, 2008). The soluble COD increase was slightly greater in the pre-treated sludge, with 4% increase with respect to the values obtained with the raw sludge. After 16 days, soluble COD in both cases starts decreasing, with this decrease more significant in the anaerobic process of pretreated sludge. This behavior shows that the pre-treatment improves not only the hydrolytic step of anaerobic digestion, but also the methanogenesis process.

Fig. 3C shows the effect of the pre-aeration step on VSS degradation under anaerobic conditions. The profile was similar to that obtained for total COD, with greater VSS removal in the process with pre-aeration. Without pre-aeration, VSS degradation was only 33%, while with pre-aeration it was 68%, or twice as large. Fig. 3C also shows that VSS removal was faster during the first 16 days, coinciding with the increase of soluble COD in the liquid phases. The greater removal obtained in the process with pre-aeration can be attributed to the solubilization of slow biodegradable organic matter in the aerobic step, thereby increasing the total removal of VSS. Greater VS removal in aerobically pretreated wastes was also reported by Xu et al. (2014), who worked with food waste, finding that hydrolysis in pretreated systems varied between 66.3% and 78.1%, showing improvement compared to the control (61.2%).

The greater removal of total COD, soluble COD and VSS was also reflected in higher methane generation in the anaerobic treatment with pre-aerated mixed sludge as seen in Fig. 4, which shows that the methane accumulated in the process with pre-aeration was 211% greater than that obtained in the process without pre-aeration. It should be noted that this increase is greater than those achieved using other pretreatments. For instance, increases between 20% and 79% have been reported with mechanical pretreatment, between 20% and 85.9% with thermal pretreatment, and between 20% and 112% with biological pretreatment (Harris and McCabe, 2015).

The methane volumes obtained in both assays are in the range of values reported in other studies. Huilinir et al. (2015), using sewage sludge from the paper and pulp industry, obtained close to 225 mL of CH_4 per gram of VS_{added}, and Xu et al. (2014) also obtained values of 270 mL of CH_4 per gram of VS_{added}. The higher methane production in the system with pretreated sludge agrees with that of other studies. Diaz et al. (2011) showed that aeration reduces the time required to reach maximum methane production,

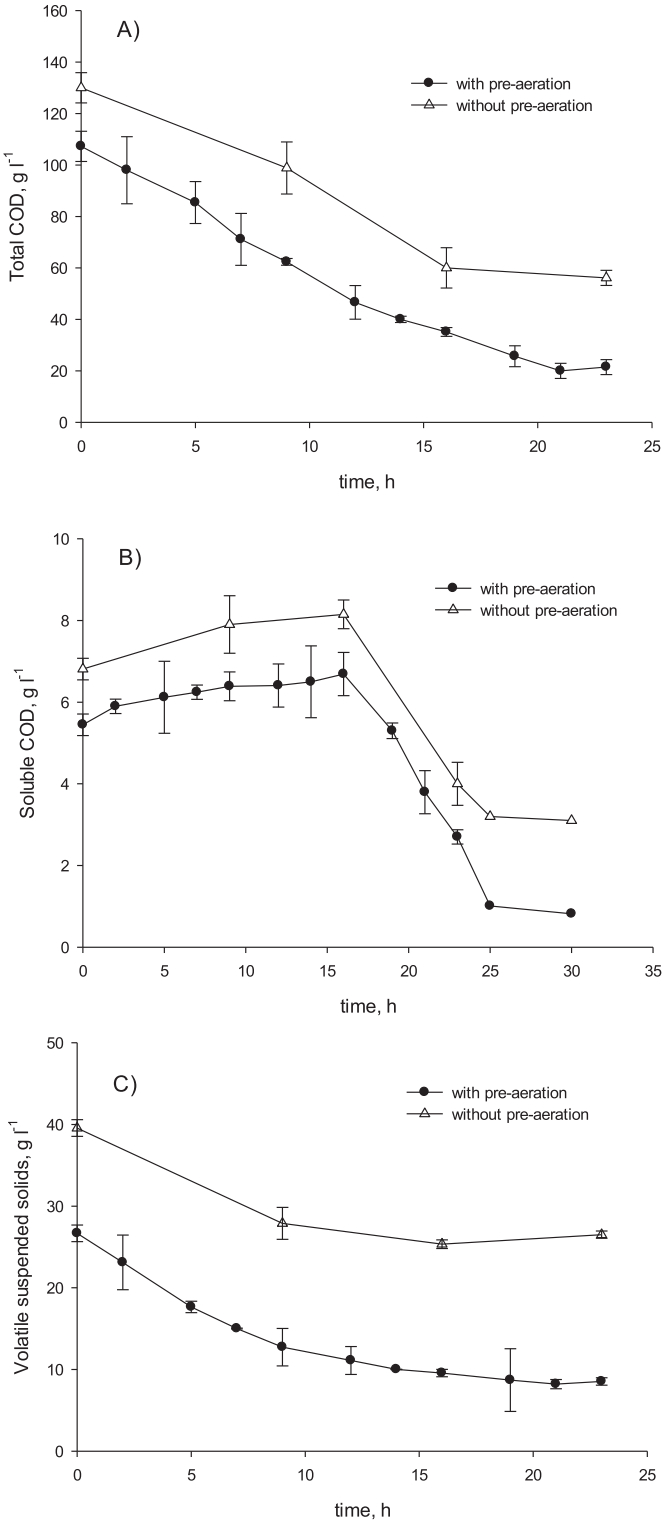


Fig. 3. Organic matter variation in the anaerobic process with and without pretreatment. A) Total COD; B) soluble COD; C) volatile solids content.

as well as its volume, in the anaerobic digestion of cellulose. Ramos and Fdz-Polanco (2013) also showed that micro-aeration contributes to increase the biogas quality in a continuous system. Xu et al. (2014) also found an increase (between 5% and 18%) in methane generation in reactors with food waste pretreated aerobically.

Thus, it is shown that pre-aeration can be used as pretreatment for increasing solubility, methane yield, and VSS removal in a batch system.

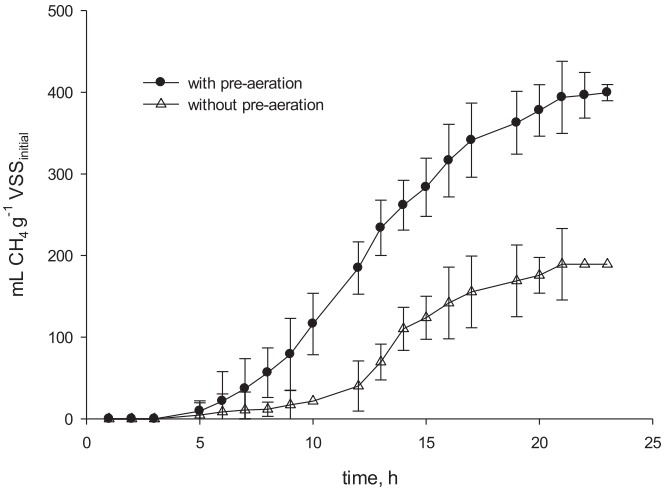


Fig. 4. Methane accumulation for pretreated and non-pretreated sewage sludge.

4. Conclusions

Pretreatment of controlled microaeration improves the hydrolytic process, with increased protein, total sugars, and soluble COD, and a greater removal of VSS compared to a system without aeration. It was determined that an air flow rate of 0.3 vvm, a 48-h aeration period, and a temperature of 35 °C were the best operational conditions for increasing the hydrolysis rate. The kinetics of VSS removal in the aerobic pretreatment follows first order kinetics, with values of k ranging between 0.226 d^{-1} and 0.269 d^{-1}. The aeration pretreatment allows a 211% increase in methane generation with respect to a process without pretreatment, improving also total COD and VSS removal.

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References

Abelleira-Pereira, J.M., Perez-Elvira, S.I., Sanchez-Oneto, J., de la Cruz, R., Portela, J.R., Nebot, E., 2015. Enhancement of methane production in mesophilic anaerobic digestion of secondary sewage sludge by advanced thermal hydrolysis pretreatment. Water Res. 71, 330–340.
APHA, AWWA, WPCF, 2012. Standard Methods for the Examination of Water and Wastewater, 22th ed. APHA, Washington, D.C.
Appels, L., Baeyens, J., Degreve, J., Dewil, R., 2008. Principles and potential of the anaerobic digestion of waste-activated sludge. Prog. Energy Combust. Sci. 34, 755–781.
Batstone, D.J., Balthes, C., Barr, K., 2010. Model assisted startup of anaerobic digesters fed with thermally hydrolysed activated sludge. Water Sci. Technol. 62, 1661–1666.
Burgess, J.E., Pletschke, B.I., 2008. Hydrolytic enzymes in sewage sludge treatment: a mini-review. Water SA 34, 343–349.
Charles, W., Walker, L., Cord-Ruwisch, R., 2009. Effect of pre-aeration and inoculum on the start-up of batch thermophilic anaerobic digestion of municipal solid waste. Bioresour. Technol. 100, 2329–2335.
Chen, Y., Cheng, J.J., Creamer, K.S., 2008. Inhibition of anaerobic digestion process: a review. Bioresour. Technol. 99, 4044–4064.
Diak, J., Ormeci, B., Kennedy, K.J., 2013. Effect of micro-aeration on anaerobic digestion of primary sludge under septic tank conditions. Bioprocess. Biosyst. Eng. 36, 417–424.
Diaz, I., Donoso-Bravo, A., Fdz-Polanco, M., 2011. Effect of microaerobic conditions on the degradation kinetics of cellulose. Bioresour. Technol. 102, 10139–10142.
Divyalakshmi, P., Murugan, D., Sivarajan, M., Sivasamy, A., Saravanan, P., Rai, C.L., 2015. In situ disruption approach on aerobic sludge biomass for excess sludge

reduction in tannery effluent treatment plant. Chem. Eng. J. 276, 130–136.
Dubois, M., Gilles, K.A., Hamilton, J.K., Reberts, P.A., Smith, F., 1956. Colorimetric method for determination of sugar and related substances. Anal. Chem. 28, 350–356.
Dumas, C., Perez, S., Paul, E., Lefebvre, X., 2010. Combined thermophilic aerobic process and conventional anaerobic digestion: effect on sludge biodegradation and methane production. Bioresour. Technol. 101, 2629–2636.
Fu, S.F., He, S., Shi, X.S., Katukuri, N.R., Dai, M., Guo, R.B., 2015. The chemical properties and microbial community characterization of the thermophilic microaerobic pretreatment process. Bioresour. Technol. 198, 497–502.
Gao, S.M., Zhao, M.X., Chen, Y., Yu, M.J., Ruan, W.Q., 2015. Tolerance response to in situ ammonia stress in a pilot-scale anaerobic digestion reactor for alleviating ammonia inhibition. Bioresour. Technol. 198, 372–379.
Gerritse, J., Schut, F., Gottschal, J.C., 1990. Mixed chemostat cultures of obligately aerobic and fermentative or methanogenic bacteria grown under oxygen-limiting conditions. FEMS Microbiol. Lett. 66, 87–89.
Goel, R., Mino, T., Satoh, H., Matsuo, T., 1998. Enzyme activities under anaerobic and aerobic conditions inactivated sludge sequencing batch reactor. Water Res. 32, 2081–2088.
Harris, P.W., McCabe, B.K., 2015. Review of pretreatments used in anaerobic digestion and their potential application in high-fat cattle slaughterhouse wastewater. Appl. Energy 155, 560–575.
Huilinir, C., Aspe, E., Roeckel, M., 2011. Modeling of the denitrification/anaerobic digestion process of salmon fishery wastewater in a biofilm tubular reactor. J. Environ. Manage. 92, 1591–1608.
Huilinir, C., Montalvo, S., Guerrero, L., 2015. Biodegradability and methane production from secondary paper and pulp sludge: effect of fly ash and modeling. Water Sci. Technol. 72, 230–237.
Huilinir, C., Quintriqueo, A., Antileo, C., Montalvo, S., 2014. Methane production from secondary paper and pulp sludge: effect of natural zeolite and modeling. Chem. Eng. J. 257, 131–137.
Jagadabhi, P.S., Kaparaju, P., Rintala, J., 2010. Effect of micro-aeration and leachate replacement on COD solubilization and VFA production during mono-digestion of grass-silage in one-stage leach-bed reactors. Bioresour. Technol. 101, 2818–2824.
Jang, H.M., Cho, H.U., Park, S.K., Ha, J.H., Park, J.M., 2014. Influence of thermophilic aerobic digestion as a sludge pretreatment and solids retention time of mesophilic anaerobic digestion on the methane production, sludge digestion and microbial communities in a sequential digestion process. Water Res. 48, 1–14.
Jenicek, P., Celis, C.A., Koubova, J., Pokorna, D., 2011. Comparison of microbial activity in anaerobic and microaerobic digesters. Water Sci. Technol. 63, 2244–2249.
Jenicek, P., Koubova, J., Bindzar, J., Zabranska, J., 2010. Advantages of anaerobic digestion of sludge in microaerobic conditions. Water Sci. Technol. 62, 427–434.
Johansen, J.E., Bakke, R., 2006. Enhancing hydrolysis with microaeration. Water Sci. Technol. 53, 43–50.
Kato, S., Haruta, S., Cui, Z.J., Ishii, M., Igarashi, Y., 2005. Stable coexistence of five bacterial strains as a cellulose-degrading community. Appl. Environ. Microbiol. 71, 7099–7106.
Kim, D., Lee, K., Park, K.Y., 2015. Enhancement of biogas production from anaerobic digestion of waste activated sludge by hydrothermal pretreatment. Int. Biodeterior. Biodegrad. 101, 42–46.
Komemoto, K., Lim, Y.G., Nagao, N., Onoue, Y., Niwa, C., Toda, T., 2009. Effect of temperature on VFA's and biogas production in anaerobic solubilization of food waste. Waste Manage. 29, 2950–2955.
Krayzelova, L., Bartacek, J., Kolesarova, N., Jenicek, P., 2014. Microaeration for hydrogen sulfide removal in UASB reactor. Bioresour. Technol. 172, 297–302.
Kruger, N.J., 1994. The Bradford method for protein quantitation. In: Methods in Molecular Biology. Humana Press, New Jersey, pp. 9–15.
Lim, J.W., Wang, J.Y., 2013. Enhanced hydrolysis and methane yield by applying microaeration pretreatment to the anaerobic co-digestion of brown water and food waste. Waste Manage 33, 813–819.
Liu, J.Y., Song, Y.M., Liu, Y.H., Ruan, R., 2015. Fungal pretreatment of effluent from piggery anaerobic digestion by Phanerochaete chrysosporium. Clean-Soil Air Water 43, 1190–1196.
Luo, K., Yang, Q., Li, X.M., Yang, G.J., Liu, Y., Wang, D.B., Zheng, W., Zeng, G.M., 2012. Hydrolysis kinetics in anaerobic digestion of waste activated sludge enhanced by alpha-amylase. Biochem. Eng. J. 62, 17–21.
Mata-Alvarez, J., Mace, S., Llabres, P., 2000. Anaerobic digestion of organic solid wastes. An overview of research achievements and perspectives. Bioresour. Technol. 74, 3–16.
Miron, Y., Zeeman, G., Van Lier, J.B., Lettinga, G., 2000. The role of sludge retention time in the hydrolysis and acidification of lipids, carbohydrates and proteins during digestion of primary sludge in CSTR systems. Water Res. 34, 1705–1713.
Montalvo, S., San Martin, J., Huilinir, C., Guerrero, L., Borja, R., 2014. Assessment of a UASB reactor with high ammonia concentrations: effect of zeolite addition on process performance. Process Biochem. 49, 2220–2227.
Morgan-Sagastume, F., Allen, D.G., 2005. Activated sludge deflocculation under temperature upshifts from 30 to 45 degrees C. Water Res. 39, 1061–1074.
Mshandete, A., Bjornsson, L., Kivaisi, A.K., Rubindamayugi, S.T., Mattiasson, B., 2005. Enhancement of anaerobic batch digestion of sisal pulp waste by mesophilic aerobic pretreatment. Water Res. 39, 1569–1575.
Mutschlechner, M., Illmer, P., Wagner, A.O., 2015. Biological pretreatment: enhancing biogas production using the highly cellulolytic fungus Trichoderma viride. Waste Manage. 43, 98–107.
Nguyen, P.H.L., Kuruparan, P., Visvanathan, C., 2007. Anaerobic digestion of municipal solid waste as a treatment prior to landfill. Bioresour. Technol. 98, 380–387.
Ramos, I., Fdz-Polanco, M., 2013. The potential of oxygen to improve the stability of anaerobic reactors during unbalanced conditions: results from a pilot-scale digester treating sewage sludge. Bioresour. Technol. 140, 80–85.
Saidu, M., Yuzir, A., Salim, M.R., Salmiati, Azman, S., Abdullah, N., 2014. Biological pretreated oil palm mesocarp fibre with cattle manure for biogas production by anaerobic digestion during acclimatization phase. Int. Biodeterior. Biodegrad. 95 (Part A), 189–194.
Vavilin, V.A., Fernandez, B., Palatsi, J., Flotats, X., 2008. Hydrolysis kinetics in anaerobic degradation of particulate organic material: an overview. Waste Manage. 28, 939–951.
Xu, S.Y., Selvam, A., Wong, J.W.C., 2014. Optimization of micro-aeration intensity in acidogenic reactor of a two-phase anaerobic digester treating food waste. Waste Manage. 34, 363–369.
Zhen, G.Y., Lu, X.Q., Li, Y.Y., Zhao, Y.C., 2014. Combined electrical-alkali pretreatment to increase the anaerobic hydrolysis rate of waste activated sludge during anaerobic digestion. Appl. Energy 128, 93–102.
Zhu, M., Lu, F., Hao, L.P., He, P.J., Shao, L.M., 2009. Regulating the hydrolysis of organic wastes by micro-aeration and effluent recirculation. Waste Manage. 29, 2042–2050.